

Long, streamlined nose cuts through the air smoothly

JRN700 Shinkansen Japan 2007

186 mph (300 km/h)

Aluminum body

LSER Class 395 Javelin UK 2009

Nose cone contains horn and coupler

140 mph (225 km/h)

Automatic sliding doors

Hyundai Rotem KTX South Korea 2010

Train's wheels are powered by eight electric motors

190 mph (305 km/h)

SNCF TGV Euroduplex France 2012

199 mph (320 km/h)

Train can seat 560 passengers

NTV AGV ETR 575 Italy 2012

186 mph (300 km/h)

Nose cone contains coupler to link train to other locomotives

IO Series Shinkansen Japan Under Development

373 mph (600 km/h)

Track-side electromagnets propel train to high speeds

In 2007, a modified TGV set a world speed record of **357 mph** (575 km/h).

electric motors housed in a power unit at the front of the train, the **DB ICE 3** has its motors spread out over the entire length of the train to distribute the weight. The **SNCF TGV Euroduplex** is a rare example of a high-speed double-decker train. Some trains use powerful

electromagnets to raise them above their track and move them along. This is called magnetic levitation (maglev). The first public passenger maglev train was the **Birmingham Airport Maglev** in the UK, while the fastest is the **Shanghai Transrapid Maglev**, in China.